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Polchinski Details
Random Examples

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Contents

CHAPTER

INTRODUCTION

PAGE 1

1.1 A First look at String Theory

1

Action Principles — 1 • The open string spectrum — 5

Chapter 1

Introduction

This Document is collection of the details from Polchinski's String Theory Vol 1 and 2 (in the context of quantum gravity) that I replicated in my own words and notation for my own understanding.

Section 1.1

A First look at String Theory

Subsection 1.1.1

Action Principles

Proposition 1.1.1

A string is a one-dimensional extended object that sweeps out a two-dimensional worldsheet in spacetime as it evolves over time. The string moves through a D-dimensional spacetime with a metric $\eta_{\mu\nu} = (-, +, + \dots, +)$

Proposition 1.1.2 Arbitrary Parameterization

$$X'^{\mu}(\tau, \sigma) = X^{\mu}(\tau', \sigma') \quad (1.1)$$

Definition 1.1.1: :

$$S_{pp} = -m \int d\tau \sqrt{-\dot{X}^{\mu} \dot{X}_{\mu}}.$$

Proposition 1.1.3

$$\delta S_{pp} = -m \int d\tau \dot{u}_\mu \delta X^\mu \quad (1.2)$$

with $u^\mu = \frac{\dot{X}^\mu}{\sqrt{-\dot{X}^\nu \dot{X}_\nu}}$

$$\delta S_{pp} = -m \int d\tau \frac{-\dot{X}_\mu \delta \dot{X}^\mu}{\sqrt{-\dot{X}^\nu \dot{X}_\nu}}.$$

IBP: $u_\mu = (-\dot{x}^\nu \dot{x}_\nu)^{-\frac{1}{2}} \dot{x}_\mu$, $dv = \frac{d}{d\tau} \delta x^\mu$

$$\delta S_{pp} = -m \left[u_\mu \delta x^\mu \right]_{\tau_i}^{\tau_f} - m \int d\tau \frac{du_\mu}{d\tau} \delta x^\mu.$$

where the boundary term vanishes since $\delta x^\mu(\tau_i) = \delta x^\mu(\tau_f) = 0$ so we verify (1.2).

Proposition 1.1.4 Equivlant form

$$S'_{pp} = \frac{1}{2} \int d\tau (\eta^{-1} \dot{X}^\mu \dot{X}_\mu - \eta m^2) \quad (1.3)$$

$$\eta(\tau) = (-\gamma_{\tau\tau}(\tau))^{1/2}.$$

Given (1.3) we can verify the form of η

$$\frac{\partial \mathcal{L}}{\partial \eta} = -(\eta)^{-2} \dot{X}^\mu \dot{X}_\mu - m^2 = 0.$$

$$\implies \eta^{-1} = \frac{\sqrt{-\dot{X}^\mu \dot{X}_\mu}}{m}.$$

$$S'_{pp} = \frac{1}{2} \int d\tau \left(\sqrt{\frac{-m^2}{\dot{X}^\mu \dot{X}_\mu}} \dot{X}^\mu \dot{X}_\mu - m \sqrt{-\dot{X}^\mu \dot{X}_\mu} \right) \quad (1.4)$$

$$= \frac{m}{2} \int d\tau \left(-\sqrt{-\dot{X}^\mu \dot{X}_\mu} - \sqrt{-\dot{X}^\mu \dot{X}_\mu} \right) = -m \int d\tau \sqrt{-\dot{X}^\mu \dot{X}_\mu} = S_{pp} \quad (1.5)$$

Definition 1.1.2: Induced Metric :

$$h_{ab} = \partial_a X^\mu(\tau, \sigma) \partial_b X_\mu(\tau, \sigma).$$

Definition 1.1.3: Nambu-Goto :

$$S_{NG} = \int_{\mathcal{M}} d\tau d\sigma \mathcal{L}_{NG}$$

$$\mathcal{L}_{NG} = -\frac{1}{2\pi\alpha'} \sqrt{-\det(h_{ab})}.$$

$\mathcal{M}$ is the worldsheet swept out by the string in spacetime.

The regge slope α' is related to the string tension T by

$$T = \frac{1}{2\pi\alpha'}.$$

Proposition 1.1.5 Symmetries of Nambu-Goto Action

- Poincaré Invariance: $X^\mu \rightarrow \Lambda^\mu_\nu X^\nu + a^\mu$
- Worldsheet Reparameterization Invariance: $(\tau, \sigma) \rightarrow (\tau', \sigma')$

Include proof XXX

Definition 1.1.4: Polyakov action :

$$S_P = -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X_\mu \quad (1.6)$$

$$\delta(\sqrt{-\gamma}) = \frac{1}{2} \sqrt{-\gamma} \gamma_{cd} \delta\gamma^{cd} \quad \delta(\gamma^{cd} \sqrt{-\gamma}) = (\delta\gamma^{cb}) \sqrt{-\gamma} + \sqrt{-\gamma} \gamma_{ad} \delta\gamma^{ad} \gamma^{cb} \cdot \frac{1}{2}.$$

$$\begin{aligned} \delta_\gamma S_P &= -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \partial_a X^\mu \partial_b X_\mu \left[(\delta\gamma^{ab}) \sqrt{-\gamma} + \sqrt{-\gamma} \gamma_{cd} \delta\gamma^{cd} \gamma^{ab} \cdot \frac{1}{2} \right] \\ &= -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \left[\partial_a X^\mu \partial_b X_\mu - \frac{1}{2} \gamma_{ab} \gamma^{cd} \partial_c X^\mu \partial_d X_\mu \right] \delta\gamma^{ab} \\ &= \frac{1}{4\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \delta\gamma^{ab} \left(h_{ab} - \frac{1}{2} \gamma^{cd} h_{cd} \gamma_{ab} \right) \\ &\implies h_{ab} = \frac{1}{2} \gamma^{cd} h_{cd} \gamma_{ab}, \quad h_{ab} (-h)^{-1/2} = \gamma_{ab} (-\gamma)^{-1/2}. \end{aligned}$$

Sub into (1.6) we get

$$S_P \rightarrow -\frac{1}{2\pi\alpha'} \int d\tau d\sigma \sqrt{-h} = S_{NG}.$$

Proposition 1.1.7 Polyakov Action Symmetries

- Poincaré Invariance (D dim): $X^\mu \rightarrow \Lambda^\mu_\nu X^\nu + a^\mu$
- Diffeomorphism Invariance: $X^\mu(\tau, \sigma) \rightarrow X^{\mu'}(\tau', \sigma')$
- Weyl Invariance (2 dim): $\gamma_{ab}(\tau, \sigma) \rightarrow e^{2\omega(\tau, \sigma)} \gamma_{ab}(\tau, \sigma)$

Prove XXX

Proposition 1.1.8 Diff invariance $\implies \vec{\nabla}_a T^{ab} = 0$

Prove XXX

Proposition 1.1.9 Equations of Motion from Polyakov Action

$$\partial_a \left(\sqrt{-\gamma} \gamma^{ab} \partial_b X^\mu \right) = 0 \quad (1.7)$$

$$S_P = -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X^\mu \partial_b X_\mu$$

$$\delta_{X^\mu} S_P = -\frac{1}{2\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \gamma^{ab} \partial_a X_\mu \partial_b \delta X^\mu$$

IBP: $u = \delta X^\mu$, $dv = \partial_b \left(\sqrt{-\gamma} \gamma^{ab} \partial_a X_\mu \right) d\sigma d\tau$.

$$\begin{aligned} \delta_{X^\mu} S_P &= -\frac{1}{2\pi\alpha'} \left[\int d\tau \sqrt{-\gamma} \gamma^{ab} \partial_a X_\mu \delta X^\mu d\sigma \right] \Big|_{\partial\mathcal{M}} + \frac{1}{2\pi\alpha'} \int d\tau d\sigma \delta X^\mu \partial_b \left(\sqrt{-\gamma} \gamma^{ab} \partial_a X_\mu \right) \\ &= 0 + \frac{1}{2\pi\alpha'} \int d\tau d\sigma \delta X^\mu \partial_b \left(\sqrt{-\gamma} \gamma^{ab} \partial_a X_\mu \right) \\ &\implies \partial_a \left(\sqrt{-\gamma} \gamma^{ab} \partial_b X^\mu \right) \delta X^\mu = 0. \end{aligned}$$

So we have two sets of EOMs corresponding to Neumann and Dirichlet boundary conditions corresponding to open and closed strings respectively.

- Neumann: $\partial_\sigma X^\mu|_{\partial\mathcal{M}} = 0$
- Dirichlet: $\delta X^\mu|_{\partial\mathcal{M}} = 0$

If we relax Poincare invariance then we can have mixed boundary conditions.

Proposition 1.1.10 Additional term for Polyakov Action (without boundary)

$$\chi = \frac{1}{4\pi} \int_{\mathcal{M}} d\tau d\sigma \sqrt{-\gamma} R \quad (1.8)$$

Prove XXX,

Subsection 1.1.2

The open string spectrum

Definition 1.1.5: Light-cone/Infinite momenta coordinates :

$$x^\pm = \frac{1}{\sqrt{2}} (x^0 \pm x^1), \quad i = 2, \dots, D-1.$$

$$a^\mu b_\mu = -a^+ b^- - a^- b^+ + a^i b^i \quad a_- = -a^+, \quad a_+ = -a^-, \quad a_i = a^i.$$

Proposition 1.1.11

If we fix our Parameterization such that $X^+(\tau) = \tau$

$$S'_\rho = \frac{1}{2} \int d\tau d\sigma \left(-2\eta^{-1} \dot{X}^- + \eta^{-1} \dot{X}^i \dot{X}^i - \eta m^2 \right) \quad (1.9)$$

Proof:

$$\dot{X}^\mu \dot{X}_\mu = -(\dot{X}^0)^2 + \left[\sum_{i=1}^{d-2} (\dot{x}^i)^2 + (\dot{X}^{D-1})^2 \right]$$

$$X^0 = \frac{1}{\sqrt{2}} (X^+ + X^-) \dot{X}^{D-1} = \frac{1}{\sqrt{2}} (X^+ - X^-)$$

$$\Rightarrow \dot{X}^\mu \dot{X}_\mu = - \left[\frac{1}{2} (x^+ + x^-)^2 \right] + \left[\frac{1}{2} (x^+ - x^-)^2 \right] + \sum_{i=1}^{D-2} (\dot{X}^i)^2$$

$$\Rightarrow \dot{X}^\mu \dot{X}_\mu = -2\dot{X}^+ \dot{X}^- + \sum_{i=1}^{D-2} (\dot{X}^i)^2$$

$$X = \tau \Rightarrow \dot{X} = 1$$

$$\Rightarrow S'_{pp} = \frac{1}{2} \int d\tau \left[\eta^{-1} (-2\dot{X}^- + \dot{X}^i \dot{X}^i) - \eta m^2 \right].$$

We arrive at (1.9).

⊙

Definition 1.1.6: Gauge Fixing Polyakov :

$$\tau = x^+, \partial_\sigma \gamma_{\sigma\sigma} = 0, \det \gamma = -1 \quad (1.10)$$

Proposition 1.1.12 Implications of Gauge Fixing

Given the identification:

$$\begin{bmatrix} \gamma^{\tau\tau} & \gamma^{\tau\sigma} \\ \gamma^{\sigma\tau} & \gamma^{\sigma\sigma} \end{bmatrix} = \begin{bmatrix} -\gamma_{\sigma\sigma}(\tau) & \gamma_{\tau\sigma}(\tau, \sigma) \\ \gamma_{\tau\sigma} & \gamma^{-1}(\tau) (1 - \gamma_{\tau\sigma}^2(\tau, \sigma)) \end{bmatrix} \quad (1.11)$$

Then the Polyakov lagrangian becomes

$$\mathcal{L} = -\frac{1}{4\pi\alpha'} \left[\gamma_{\sigma\sigma} \left(2\partial_\tau X^- - \partial_\tau X^i \partial_\tau X^i \right) - 2\gamma_{\sigma\tau} \left(\partial_\sigma X^- - \partial_\tau X^i \partial_\sigma X^i \right) + \gamma_{\sigma\sigma}^{-1} (1 - \gamma_{\tau\sigma}^2) \partial_\sigma X^i \partial_\sigma X^i \right] \quad (1.12)$$

Proof:

$$\begin{aligned} & \gamma^{ab} \left(-2\partial_a X^+ \partial_b X^- + \partial_a X^i \partial_b X^i \right) \\ &= -2\gamma^{\tau\tau} \partial_\tau X^- + \gamma^{\tau\tau} \partial_\tau X^i \partial_\tau X^i + \left(-2\gamma^{\sigma\tau} \partial_\sigma X^+ \partial_\tau X^- + \gamma^{\sigma\tau} \partial_\sigma X^i \partial_\tau X^i \right) \\ &+ \left(-2\gamma^{\tau\sigma} \partial_\sigma X^- + \gamma^{\tau\sigma} \partial_\tau X^i \partial_\sigma X^i \right) + \left(\gamma^{\sigma\sigma} \partial_\sigma X^i \partial_\sigma X^i \right) \\ &= \gamma_{\sigma\sigma} \left(2\partial_\tau X^- - \partial_\tau X^i \partial_\tau X^i \right) - 2\gamma_{\sigma\tau} \left(-\partial_\sigma X^+ \partial_\tau X^- + \partial_\sigma X^- \right) \\ &+ \left(\gamma_{\sigma\sigma}^{-1} (1 - \gamma_{\tau\sigma}^2) \partial_\sigma X^i \partial_\sigma X^i \right). \end{aligned}$$

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Definition 1.1.7: Mean position and zero mean :

$$x^-(\tau) = \frac{1}{l} \int_0^l d\sigma X^-(\tau, \sigma) \quad (1.13)$$

$$Y^-(\tau, \sigma) = X^-(\tau, \sigma) - x^-(\tau) \quad (1.14)$$

Proposition 1.1.13 Modified boundary conditions

Under the gauge fixing (1.10) and definitions (1.13), (1.14) the boundary conditions become

$$\gamma_{\tau\sigma} \partial_\tau X^\mu - \gamma_{\tau\tau} \partial_\sigma X^\mu = 0 \quad \text{at } \sigma = 0, l \quad (1.15)$$

Proof:

$$\begin{aligned} & \partial^\sigma X^\mu(\tau, \sigma) = \partial^\sigma X^\mu(\tau, l) = 0 \\ \mu = + & \quad X^+ = \tau \quad \partial_\tau = 1, \quad \partial_\sigma X^+ = 0 \\ \Rightarrow & \quad \gamma^{\sigma\tau} \partial_\tau X^+ - \gamma^{\sigma\sigma} \partial_\sigma X^+ = \gamma^{\sigma\tau} \Big|_{\sigma=0}^{\sigma=l} = 0 \\ \mu = i & \\ \Rightarrow & \quad \gamma^{\sigma\sigma} \partial_\sigma X^i + \gamma^{\sigma\tau} \partial_\tau X^i = \gamma^{\sigma\sigma} \partial_\sigma X^i = 0 \quad \partial_\sigma X^i \Big|_{\sigma=0}^{\sigma=l} = 0 \end{aligned}$$

☺

Proposition 1.1.14 Form of $\mathcal{L}$ after gauge and lagrange multiplier

$$\mathcal{L} = -\frac{l}{2\pi\alpha'}\gamma_{\sigma\sigma}\partial_\tau x^- + \frac{1}{4\pi\alpha'} \int_0^l d\sigma \left(\gamma_{\sigma\sigma}\partial_\tau X^i \partial_\tau X^i - \gamma_{\sigma\sigma}^{-1}\partial_\sigma X^i \partial_\sigma X^i \right) \quad (1.16)$$

Proof:

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Proposition 1.1.15 Form of momenta & momenta density

$$p_- = -p^+ = -\frac{l}{2\pi\alpha'}\gamma_{\sigma\sigma} \quad (1.17)$$

$$\Pi^i = \frac{1}{2\pi\alpha'}\gamma_{\sigma\sigma}\partial_\tau X^i = -\frac{p^+}{l}\partial_\tau X^i \quad (1.18)$$

Proof:

$$\begin{aligned} \mathcal{L} &= -\frac{l}{2\pi\alpha'}\gamma_{\sigma\sigma}\partial_\tau x^- + \frac{1}{4\pi\alpha'} \int_0^l d\sigma \left(\gamma_{\sigma\sigma}\partial_\tau X^i \partial_\tau X^i - \gamma_{\sigma\sigma}^{-1}\partial_\sigma X^i \partial_\sigma X^i \right) \\ p_- &= \frac{\partial \mathcal{L}}{\partial (\partial_\tau x^-)} = -\frac{l}{2\pi\alpha'}\gamma_{\sigma\sigma} \\ \Pi^i &= \frac{\delta \mathcal{L}}{\delta (\partial_\tau X^i)} = 0 + \frac{1}{4\pi\alpha'} \int_0^l d\sigma \frac{\delta}{\delta (\partial_\tau X^i)} \left(\gamma_{\sigma\sigma}\partial_\tau X^i \partial_\tau X^i - \gamma_{\sigma\sigma}^{-1}\partial_\sigma X^i \partial_\sigma X^i \right) \\ \text{use } \frac{\delta (\partial_\tau X^i (\sigma'))}{\delta (\partial_\tau X^i (\sigma))} &= \delta^{ij} \delta (\sigma - \sigma') \\ \Pi^i &= \frac{1}{4\pi\alpha'} \int_0^l d\sigma \left[\gamma_{\tau\sigma}\partial_\tau x^i \delta^{ij} (\sigma - \sigma') + \gamma_{\tau\sigma}\delta^{ij} (\sigma - \sigma') \partial_\tau X^i \right] \\ \Pi^i &= \frac{1}{2\pi\alpha'}\gamma_{\sigma\sigma}\partial_\tau X^i \quad -p^+ = p_- = -\frac{l}{2\pi\alpha'}\gamma_{\sigma\sigma} \\ \Rightarrow \Pi^i &= -\frac{p^+}{l}\partial_\tau X^i. \end{aligned}$$

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Proposition 1.1.16 Hamiltonian for $D - 2$ free fields

$$\mathcal{H} = \frac{1}{4\pi\alpha'p^+} \int_0^l d\sigma \left(2\pi\alpha'\Pi^i\Pi^i + \frac{1}{2\pi\alpha'}\partial_\sigma X^i \partial_\sigma X^i \right) \quad (1.19)$$

Proof:

$$\begin{aligned}
\mathcal{H} &= p_- \partial_\tau x^- - \mathcal{L} + \int_0^l d\sigma \Pi^i \partial_\tau X^i \\
\mathcal{H} &= \left[-\frac{l}{2\pi\alpha'} \gamma_{\sigma\sigma} \right] \partial_\tau \left[\frac{1}{l} \int_0^l d\sigma X^-(\tau, \sigma) \right] \\
&\quad - \left[-\frac{l}{2\pi\alpha'} \gamma_{\sigma\sigma} \partial_\tau x^- + \frac{1}{4\pi\alpha'} \int_0^l d\sigma \left(\gamma_{\sigma\sigma} \partial_\tau X^i \partial_\tau X^i - \gamma_{\sigma\sigma}^{-1} \partial_\sigma X^i \partial_\sigma X^i \right) \right] \\
&\quad + \int_0^l d\sigma \left(-\frac{p^+}{l} \partial_\tau X^i \right) \partial_\tau X^i \\
\mathcal{H} &= \frac{1}{4\pi\alpha'} \int_0^l d\sigma \left(-\gamma_{\sigma\sigma} \partial_\tau X^i \partial_\tau X^i + \gamma_{\sigma\sigma}^{-1} \partial_\sigma X^i \partial_\sigma X^i \right) - \frac{p^+}{l} \int_0^l d\sigma \partial_\tau X^i \partial_\tau X^i \\
\text{Sub (1.18)} \implies \mathcal{H} &= \frac{1}{4\pi\alpha' p^+} \int_0^l d\sigma \left(2\pi\alpha' \Pi^i \Pi^i + \frac{1}{2\pi\alpha'} \partial_\sigma X^i \partial_\sigma X^i \right).
\end{aligned}$$

⊙

Proposition 1.1.17 Form of Hamiltons equations

$$\partial_\tau x^- = \frac{\partial \mathcal{H}}{\partial p_-} = \frac{\mathcal{H}}{p^+}, \quad \partial_\tau p^+ = -\frac{\partial \mathcal{H}}{\partial x^-} = 0 \quad (1.20)$$

$$\partial_\tau X^i = \frac{\delta \mathcal{H}}{\delta \Pi^i} = 2\pi\alpha' c \Pi^i, \quad \partial_\tau \Pi^i = -\frac{\delta \mathcal{H}}{\delta X^i} = \frac{c}{2\pi\alpha'} \partial_\sigma \partial_\sigma X^i \quad (1.21)$$

$$\text{where } c = \frac{l}{2\pi\alpha' p^+}.$$

Proof:

(1.20) is straightforward from (1.19). For (1.21) we have the second term in

$\mathcal{H}$

$$\begin{aligned}
\mathcal{I} &= \frac{1}{8\pi^2 (\alpha')^2 p^+} \int_0^l d\sigma \partial_\sigma X^i \partial_\sigma X^i \\
\delta \mathcal{I} &= \frac{1}{4\pi^2 (\alpha')^2 p^+} \int_0^l d\sigma \partial_\sigma X^i \partial_\sigma \delta X^i \\
\text{IBP } u &= \partial_\sigma X^i, \quad dv = \partial_\sigma \delta X^i d\sigma \\
\delta \mathcal{I} &= \frac{1}{4\pi^2 (\alpha')^2 p^+} \left[\partial_\sigma X^i \delta X^i \Big|_{\sigma=0}^{\sigma=l} - \int_0^l d\sigma \partial_\sigma \partial_\sigma X^i \delta X^i \right] \\
&= -\frac{1}{4\pi^2 (\alpha')^2 p^+} \int_0^l d\sigma \partial_\sigma \partial_\sigma X^i \delta X^i \\
\frac{\delta \mathcal{I}}{\delta X^i} &= -\frac{l}{4\pi^2 (\alpha')^2 p^+} \partial_\sigma \partial_\sigma X^i = -\frac{c}{2\pi\alpha'} \partial_\sigma \partial_\sigma X^i \\
\Rightarrow \partial_\tau \Pi^i &= -\frac{\delta \mathcal{H}}{\delta X^i} = \frac{c}{2\pi\alpha'} \partial_\sigma \partial_\sigma X^i.
\end{aligned}$$

For the first equation in (1.21) we have

$$\begin{aligned}
\partial_\tau X^i &= \frac{\delta \mathcal{H}}{\delta \Pi^i} = \frac{1}{4\pi\alpha' p^+} \int_0^l d\sigma 4\pi\alpha' \Pi^i \delta \Pi^i \\
&= \frac{1}{p^+} \int_0^l d\sigma \Pi^i \delta \Pi^i \\
&= \frac{l}{2\pi\alpha' p^+} 2\pi\alpha' \Pi^i = 2\pi\alpha' c \Pi^i.
\end{aligned}$$

Putting these together we get the wave equation and we have boundary conditions from before.

$$\partial_\tau \partial_\tau X^i = c^2 \partial_\sigma \partial_\sigma X^i \quad \partial_\sigma X^i \Big|_{\sigma=0}^{\sigma=l} = 0 \quad (1.22)$$

☺

Proposition 1.1.18 General solution to wave equation

$$X^i(\tau, \sigma) = X_L^i(\tau + c\sigma) + X_R^i(\tau - c\sigma) \quad (1.23)$$

$$X^i(\tau, \sigma) = x^i + \frac{p^i}{p^+} + i(2\alpha')^{1/2} \sum_{n \neq 0} \frac{1}{n} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \quad (1.24)$$

Proof:

Spacial part:

$$f_0(\sigma) = 1, \quad f_n(\sigma) = \cos(n\pi\sigma/l), \quad n \in \mathbb{Z}^+.$$

Temporal part:

$$\partial_\tau \partial_\tau x_n + \omega_n^2 x_n = 0, \quad \omega_n = \frac{n\pi c}{l}.$$

$$\implies x_n(\tau) = A_n e^{-i\omega_n \tau} + A_{-n} e^{i\omega_n \tau}.$$

Zero mode:

$$\partial_\tau \partial_\tau x_0 = 0 \implies x_0(\tau) = x^i + \frac{p^i}{p^+} \tau.$$

We can combine A_n, A_{-n} into a single complex coefficient α_n^i such that $A_n = i(2\alpha')^{1/2} \frac{1}{2} \alpha_n^i$, $A_{-n} = -i(2\alpha')^{1/2} \frac{1}{2} \alpha_{-n}^i$. Substituting back we get (1.24). ☺

Note:-

We can also find Π^i from (1.18) and (1.24):

$$\begin{aligned} \Pi^i(\tau, \sigma) &= -\frac{p^+}{l} \partial_\tau X^i(\tau, \sigma) \\ &= -\frac{p^+}{l} \left[\frac{p^i}{p^+} + (2\alpha')^{1/2} \sum_{n \neq 0} \alpha_n^i e^{-in\pi c \tau / l} \cos(n\pi \sigma / l) \right]. \end{aligned}$$

Definition 1.1.8: COM Variables :

$$x^i(\tau) = \frac{1}{l} \int_0^l d\sigma X^i(\tau, \sigma) \quad (1.25)$$

$$p^i(\tau) = \int_0^l d\sigma \Pi^i(\tau, \sigma) = \frac{p^+}{l} \int_0^l d\sigma \partial_\tau X^i(\tau, \sigma) \quad (1.26)$$

Definition 1.1.9: Equal time canonical commutation relations :

$$[x^i, p^j] = i\delta^{ij} \quad (1.27)$$

$$[X^i(\tau, \sigma), \Pi^j(\tau, \sigma')] = i\delta^{ij} \delta(\sigma - \sigma') \quad (1.28)$$

Proposition 1.1.19 Fourier coefficient commutation relations

$$[\alpha_m^i, \alpha_n^j] = n\delta^{ij} \delta_{m, -n} \quad (1.29)$$

$$[x^i, p^i] = i\delta^{ij} \quad (1.30)$$

Proof:

$$LHS = \int_0^l \cos\left(\frac{\pi m \sigma}{l}\right) \cos\left(\frac{\pi n \sigma'}{l}\right) d\sigma d\sigma' [X^i(\tau, \sigma), \Pi^j(\tau, \sigma')]$$

$$RHS = \int_0^l \cos\left(\frac{\pi m \sigma}{l}\right) \cos\left(\frac{\pi n \sigma'}{l}\right) d\sigma d\sigma' i \delta^{ij} \delta(\sigma - \sigma')$$

$$RHS = i \delta^{ij} \int_0^l \cos\left(\frac{\pi m \sigma}{l}\right) \cos\left(\frac{\pi n \sigma}{l}\right) d\sigma = i \delta^{ij} \frac{l}{2} \delta_{m, -n}$$

$$LHS = \int_0^l \int_0^l d\sigma d\sigma' \cos\left(\frac{\pi m \sigma}{l}\right) \cos\left(\frac{\pi n \sigma'}{l}\right) C^{ij}$$

$$C^{ij} = [X^i(\tau, \sigma), \Pi^j(\tau, \sigma')]$$

$$C^{ij} = \left[x^i + \frac{p^i}{p^+} + i(2\alpha')^{1/2} \sum_{n \neq 0} \frac{1}{n} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right] \cdot$$

$$\cdot \left[-\frac{p^+}{l} \left[\frac{p^i}{p^+} + (2\alpha')^{1/2} \sum_{n \neq 0} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right] \right]$$

$$- \left[\frac{p^i}{p^+} + (2\alpha')^{1/2} \sum_{n \neq 0} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right] \cdot$$

$$\cdot \left[x^i + \frac{p^i}{p^+} + i(2\alpha')^{1/2} \sum_{n \neq 0} \frac{1}{n} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right]$$

$$C^{ij} = \sqrt{2\alpha'} i \cdot \frac{1}{\sqrt{2\alpha'} l} \cdot \sum_{k, r \in \mathbb{Z}/\{0\}} \frac{1}{k} [a_k^i, a_r^j] \int_0^l \int_0^l d\sigma d\sigma' \cos\left(\frac{\pi n \sigma}{l}\right) \cos\left(\frac{\pi m \sigma'}{l}\right) \cdot e^{-i(k+r)\pi c\tau/l}$$

$$\text{Using } \underbrace{\int_0^l d\sigma \cos\left(\frac{\pi k \sigma}{l}\right) \cos\left(\frac{\pi m \sigma}{l}\right)}_{\delta_{k, m} \cdot \frac{l}{2}} \underbrace{\int_0^l d\sigma' \cos\left(\frac{\pi r \sigma'}{l}\right) \cos\left(\frac{\pi n \sigma'}{l}\right)}_{\delta_{r, n} \cdot \frac{l}{2}} = \frac{l^2}{4} \delta_{n, r} \delta_{k, m} \quad (1.31)$$

$$LHS = \frac{l^2}{4} \cdot i \cdot \frac{1}{m} \frac{1}{l} [\alpha_m^i, \alpha_n^j] e^{-i(m+n)\pi c\tau/l}, \quad \text{Pick } \tau = 0 \text{ to remove the phase}$$

$$LHS = RHS \implies [\alpha_m^i, \alpha_n^j] = m \delta^{ij} \delta_{m, -n}.$$

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Proposition 1.1.20 Normalisation

$$\alpha_m^i \sim \sqrt{m}a, \quad \alpha_{-m}^i \sim \sqrt{m}a^\dagger \quad (1.32)$$

Proof:



Proposition 1.1.21 General state

$$|N; k\rangle = \prod_{i=2}^{D-1} \prod_{n=1}^{\infty} \frac{(\alpha_{-n}^i)^{N_{in}}}{(n^{N_{in}} N_{in}!)^{\frac{1}{2}}} |0; k\rangle \quad (1.33)$$

Proof:



Proposition 1.1.22 Hamiltonian in terms creation/annihilation operators

$$\mathcal{H} = \frac{1}{2p^+} \left(p^i p^i + \frac{1}{\alpha'} \sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i + A \right) \quad (1.34)$$

Proof:

Using (1.19) and (1.24) we have

$$\begin{aligned}\mathcal{H} &= \frac{1}{4\pi\alpha'p^+} \int_0^l d\sigma \left(2\pi\alpha' \Pi^i \Pi^i + \frac{1}{2\pi\alpha'} \partial_\sigma X^i \partial_\sigma X^i \right) \\ X^i(\tau, \sigma) &= x^i + \frac{p^i}{p^+} + i(2\alpha')^{1/2} \sum_{n \neq 0} \frac{1}{n} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \\ \partial_\sigma X^i &= -i(2\alpha')^{1/2} \sum_{n \neq 0} \frac{1}{n} \alpha_n^i e^{-in\pi c\tau/l} \frac{n\pi}{l} \sin(n\pi\sigma/l) \\ \Pi^i &= -\frac{p^+}{l} \left[\frac{p^i}{p^+} + (2\alpha')^{1/2} \sum_{n \neq 0} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right] \\ \mathcal{H} &\sim \underbrace{\int_0^l \Pi^2 d\sigma}_{\mathcal{I}_1} + \underbrace{\int_0^l (\partial_\sigma X)^2 d\sigma}_{\mathcal{I}_2} \\ \mathcal{I}_1 &= \frac{p_i^2}{l} + \int_0^l \frac{p^i}{p^+} \sum_{n \neq 0} \left(\cos\left(\frac{n\pi\sigma}{l}\right) \tilde{C} \right) d\sigma + \int_0^l \left((2\alpha')^{1/2} \sum_{n \neq 0} \alpha_n^i e^{-in\pi c\tau/l} \cos(n\pi\sigma/l) \right)^2 d\sigma \\ &\quad \text{2nd term vanishes by orthogonality, 3rd term we use Parseval's identity.} \\ \mathcal{I}_1 &= \frac{p_i^2}{l} + \frac{1}{4l\alpha'} \sum_{n \neq 0} \alpha_n^i \alpha_n^i \\ \mathcal{I}_2 &= \int_0^l \left(\partial_\sigma X^i \right)^2 d\sigma = \alpha' \frac{\pi^2}{l} \sum_{n \geq 1} \alpha_n^i \alpha_{-n}^i \\ \mathcal{H} &= \frac{p_i^2}{2p^+} + \frac{1}{2p^+\alpha'} \sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i + A.\end{aligned}$$

Note:-

Parseval's identity:

Given

$$f(\sigma) = a_0 + \sum_{n \geq 1} a_n \cos(n\pi\sigma/l), \quad g(\sigma) = b_0 + \sum_{n \geq 1} b_n \cos(n\pi\sigma/l).$$

then

$$\int_0^l f \cdot g d\sigma = a_0 b_0 + \frac{l}{2} \sum_{n \geq 1} a_n b_n.$$

If we set $f = g$ and $a_0 = 0$ we have the case used above.

